

Professional experience

- Novembre 2020 - **Research engineer, INRAE GenPhySE**, Toulouse, France.
current Member of the Dynagen group and Sigenae platform
- Functional annotation of animal genomes (associated with the FAANG consortium)
 - MicroRNA annotation and characterization
 - Genotyping by sequencing for animal populations
 - Bioinformatic and biostatistic support for -omics data analysis
- January 2020 - **Research engineer, INRAE, Unités LPGP et IGEPP**, Rennes, France.
July 2020
- Microtranscriptome annotation and biomarker identification for the rainbow trout
 - Detection of large structural variants for the downy mildew genome
- January 2019 - **Research engineer, iDiv, Bioinformatics Unit**, Leipzig, Germany.
December 2019
- Member of the bioinformatics support group (BIU)
 - Assembly of highly heterozygous genomes using third generation data (Pacbio - 10X)
 - Transcriptomics
 - Population genomics (RAD-seq)
- 2015-2018 **PhD student, INRA - INRIA team Genscale**, Rennes, France.
Development and application of bioinformatics tools for insect metagenomics
- Analysis of multi-scale genomic diversity in large metagenomic datasets
 - Development of MinYS, a tool for targetted genome assembly from metagenomic data
- February 2015 – **Statistics and Bioinformatics intern, CNRS Lyon, supervised by Franck Picard**,
August 2015 Functional Poisson regression applied to peak detection in Chip-Seq NGS data.
- September 2013 - **Bioinformatics intern, Rothamsted Research**, Harpenden, United Kingdom,
February 2014 Setting up a Galaxy pipeline to represent gene abundances in metagenomic datasets.

Education

- 2018 **PhD in Bioinformatics, University of Rennes**, Rennes, France.
Development and application of bioinformatics tools for insect metagenomics. Supervisors :
- Jean-Christophe Simon (INRA, IGEPP)
 - Claire Lemaitre (Inria, IRISA)
 - Christophe Mougél (INRA, IGEPP)
- 2012–2015 **Diplôme d'ingénieur agronome (MsC in life sciences), Agrocampus Ouest**, Rennes, France.
Specializing in applied statistics for 2 years
- 2010–2012 **Classe préparatoire, Lycée Chateaubriand**, Rennes, France.
2010 **Baccalauréat, Lycée Saint Charles**, Saint Brieuc, France, with honours.

Teaching and supervising

Teaching

- 2019 **Master 2**, Molecular evolution course.
8h of practical work : assembly, alignment, phylogeny
- 2016 **Master 1**, Simulation for biostatistics.
In charge of a 64h course of statistics and programming for medicine students.
R programming, random variable generation, stochastic modelling

- 2017 **Master 1**, Statistical learnings.
Supervising 32h of statistics practical work.
Statistics and introduction to machine learning
Supervising
- 2026 **Supervising**, *Lionel Viguié (Master Biologie des systèmes, Université de Toulouse)*.
Genotype imputation using pedigree information
- 2026 **Supervising**, *N'Guessan Samuel N'GBAIN (Master Biologie des systèmes, Université de Toulouse)*.
Characterization of porcine retrocopies with functional omic data
- 2024 **Supervising**, *Océane Carpentier (Master BIG, Université de Rennes 1)*.
Multi omics intergration to discover expression regulation patterns
- 2022 **Supervising**, *Martin Racoupeau (Master bioinformatique, Université de Toulouse)*.
Comparing miRNA annotation tools
- 2017 **Supervising**, *Wesley Delage (Master BIG, Université de Rennes 1)*.
Development of an innovative method based on a reference guided assembly for the genome reconstruction of a bacteriophage
- Other**
- 2018 **Volunteer**, in RECOMB 2018 organization, Paris.
- 2024 **Volunteer**, in JOBIM 2024 organization, Toulouse.
- 2016 **Scientific dissemination**, 2nd jury prize of Sciences en court(s).

Languages

French	Native speaker	English	Proficient
German	Intermediate		

Skills

Bioinformatics and NGS : Assembly, variant calling, high volume datasets, metagenomics...

Programming	R: ★★★★★ Python: ★★★★★ Nextflow: ★★★★★ Julia: ★★★★★ C++: ★★★★★
Software	Developer of bioinformatics software (MinYS) and pipelines (TAGADA, PARSEC, pegASE)
Statistics	Inferential and descriptive statistics, data visualization
Other	Bash, Git, Latex, software distribution (Conda, Singularity, NextFlow) Proficient with Windows and GNU/Linux, including computing clusters

Selection of publications

Peer reviewed publications

Björn E Langer, Andreia Amaral, Marie-Odile Baudement, Franziska Bonath, Mathieu Charles, Praveen Krishna Chitneedi, Emily L Clark, Paolo Di Tommaso, Sarah Djebali, Philip A Ewels, et al. Empowering bioinformatics communities with nextflow and nf-core. *Genome biology*, 26(1):228, 2025.

Fanny Mollandin, Hervé Acloque, Maria Ballester, Marco Bink, Mario Calus, Daniel Crespo-Piazuelo, Pascal Croiseau, Sarah Djebali, Sylvain Foissac, Hélène Gilbert, Elisabetta Giuffra, **Cervin Guyomar**, Ole Madsen, Marie-José Mercat, Bruno da Costa Perez, Jani de Vos, and Andrea Rau. Guiding eqtl mapping and genomic prediction of gene expression in three pig breeds with tissue-specific epigenetic annotations from early development. *Genomics*, 118(1):111158, 2026.

Cyril Kurylo, **Guyomar, Cervin**, Sylvain Foissac, and Sarah Djebali. TAGADA: a scalable pipeline to improve genome annotations with RNA-seq data. *NAR Genomics and Bioinformatics*, 5(4):lqad089, October 2023.

Thomas Desvignes, Philippe Bardou, Jérôme Montfort, Jason Sydes, **Guyomar, Cervin**, Simon George, John H Pos tlethwait, and Julien Bobe. FishmiRNA: An evolutionarily supported microRNA annotation

34 rue Claude de Forbin – 31400 – Toulouse

☎ +336 65 13 37 63 • ✉ cervin.guyomar@inrae.fr

🌐 <https://cguyomar.github.io/>

and expression database for ray-finned fishes. *Molecular Biology and Evolution*, 39(2):msac004, January 2022. Publisher: Oxford University Press (OUP).

Guyomar, Cervin, Wesley Delage, Fabrice Legeai, Christophe Mougel, Jean-Christophe Simon, and Claire Lemaitre. MinYS: mine your symbiont by targeted genome assembly in symbiotic communities. *NAR Genomics and Bioinformatics*, 2(3):1–11, September 2020.

Guyomar, Cervin, Fabrice Legeai, Emmanuelle Jousselin, Christophe Mougel, Claire Lemaitre, and Jean-Christophe Simon. Multi-scale characterization of symbiont diversity in the pea aphid complex through metagenomic approaches. *Microbiome*, 6(1):181, December 2018.

Book chapters

Cervin Guyomar and Claire Lemaitre. Métagénomique et métatranscriptomique. In *Du texte aux graphes : méthodes discrètes pour la bioinformatique*. Éditions ISTE, 2020.